

Seming Rai

567-317-4479

srai14@rockets.utoledo.edu

[linkedin.com/in/seming](https://www.linkedin.com/in/seming)

semingrai.github.io/MeinMein.Portfolio

EDUCATION

University of Toledo — B.S. in Data Science, Minor in Computer Science & Engineering

Expected May 2028 | GPA: 3.82 / 4.0

Honors: Dean's List (Fall 2025)

TECHNICAL SKILLS

- Programming & Data: Python, Pandas, NumPy, Scikit-learn (Random Forest, Decision Trees, KNN)
 - Visualization & Dashboards: Streamlit, Plotly, Matplotlib, Seaborn
 - Tools & Platforms: Git/GitHub, Jupyter Notebook
-

PROJECTS

Instagram User Analytics Dashboard – *Python, Streamlit, Pandas, Plotly*

- Processed 1.2M-row Kaggle dataset, sampling 50K users for fast and accurate analysis.
- Built 8+ interactive Streamlit visualizations for user demographics, engagement, and content trends.
- Created dynamic Plotly charts analyzing gender, country, relationships, and activity metrics (sessions, reels, stories, likes, DMs).
- Optimized dashboard by sampling 1.2M-row dataset for real-time interactive exploration.

Stroke Prediction Using Machine Learning – *Python, Scikit-learn, Pandas, NumPy*

- Developed binary classification model predicting stroke risk with 94.8% accuracy on 5,110 patient records.
- Handled missing data using KNN Imputation for 201 BMI values, preserving feature relationships.
- Compared Decision Tree (90.5%) vs Random Forest (94.8%), improving accuracy by 4.3%.
- Built complete preprocessing pipeline handling missing values, categorical encoding, and train-test stratification.

Ames Housing Price Prediction – *Python, Scikit-learn, Pandas*

- Built Random Forest regression model predicting Iowa home prices with 9.5% error rate (\$17,150 MAE).
 - Processed 80+ features including 43 categorical variables with OneHot Encoding for 1,460 properties.
 - Reduced prediction error from 26% to 9.5% via full feature engineering and proper train-test splitting.
 - Improved model accuracy by 16.5 percentage points through systematic feature inclusion and encoding optimization.
-

EXPERIENCE

IT Lab Assistant (Info Coms) — University of Toledo

Dec 2025 – Present

- Provided technical support for lab systems and student workstations, troubleshooting hardware/software issues.
 - Assisted with setup, maintenance, and optimization of lab infrastructure.
 - Gained hands-on experience with academic IT workflows and user support.
-

CAMPUS & COMMUNITY INVOLVEMENT

- Runner-up, USCF U1000 Canton Chess Tournament
 - Outstanding People's Choice Award, Fall 2025 Showcase
 - Model United Nations (MUN) Candidate
 - AI/ML Club Member
 - UToledo Data Club Member
 - JuiceHouse Chess Blend Member
-

CERTIFICATES

- CS50P – Harvard University
- Kaggle Intro to Machine Learning
- Kaggle Pandas